Decision Making and Documentation

Introduction

- Decision-making is an integral part of any business no matter how small, simple or big and complex it may be. Thus decisions have to be made and set procedures are to be followed as the subsequent actions.

- Thus while analyzing and designing a business system, analyst also needs to identify and document any decision policies or business rules of the system being designed. There are various tools and techniques available to the analyst for this, like decision trees, decision tables and structured English.

- To analyze procedures and decisions the first step to be taken is to identify conditions and actions of all possible activities. Conditions are the possibilities or the possible states of any given entity, which can be a person, place, thing, or any event. Conditions are always in a flux i.e. they keep on varying time to time and object to object and based only on these conditions the decisions are made therefore conditions are also put as decision variables.
- Documentation comes as an aid in this condition based decision process. As the whole web of all the possible combination of conditions and decisions is usually very large and cumbersome it becomes extremely important to document these so that there are no mistakes committed during the decision process.

- Here comes the role of documenting tools, which are available to the analyst. Tools, which are usually used, are
    - Decision trees
    - Decision tables
    - Structured English and
    - The various CASE tools such as DFDs.
- The basic role of these tools is to depict the various conditions, their possible combinations and the subsequent decisions. This has to be done without harming the logical structure involved. Once all of the parameters are objectively represented the decision process becomes much simpler, straightforward and almost error free.

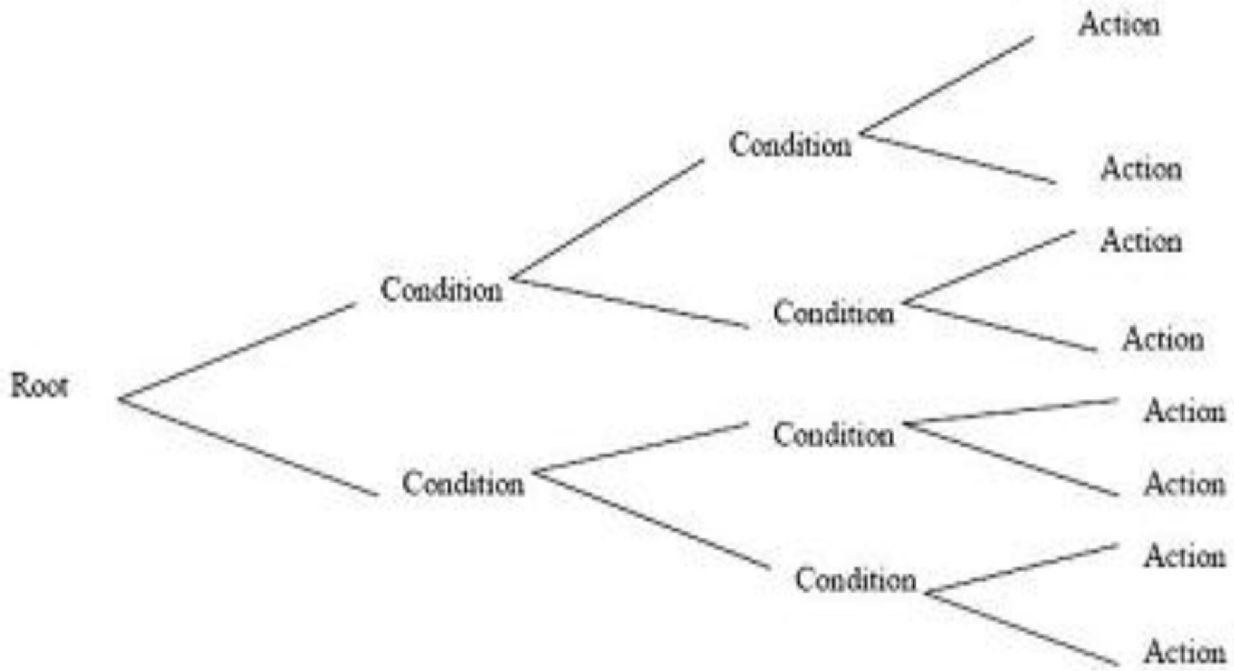
1. Decision Trees

- Decision tree is a tree like structure that represents the various conditions and the subsequent possible actions. It also shows the priority in which the conditions are to be tested or addressed. Each of its branches stands for any one of the logical alternatives and because of the branch structure, it is known as a tree.
- The decision sequence starts from the root of the tree that is usually on the left of the diagram. The path to be followed to traverse the branches is decided by the priority of the conditions and the respectable actions. A series of decisions are taken, as the branches are traversed from left to right.
- The nodes are the decision junctions. After each decision point there are next set of decisions to be considered. Therefore at every node of the tree represented conditions are considered to determine which condition prevails before moving further on the path.

Decision tree CONTD

- This decision tree representation form is very beneficial to the analyst.
  - The first advantage is that by using this form the analyst is able to depict all the given parameters in a logical format which enables the simplification of the whole decision process as now there is a very remote chance of committing an error in the decision process as all the options are clearly specified in one of the most simplest manner.
  - Secondly it also aids the analyst about those decisions, which can only be taken when couple or more conditions should hold true together for there may be a case where other conditions are relevant only if one basic condition holds true.

Decision tree CONTD

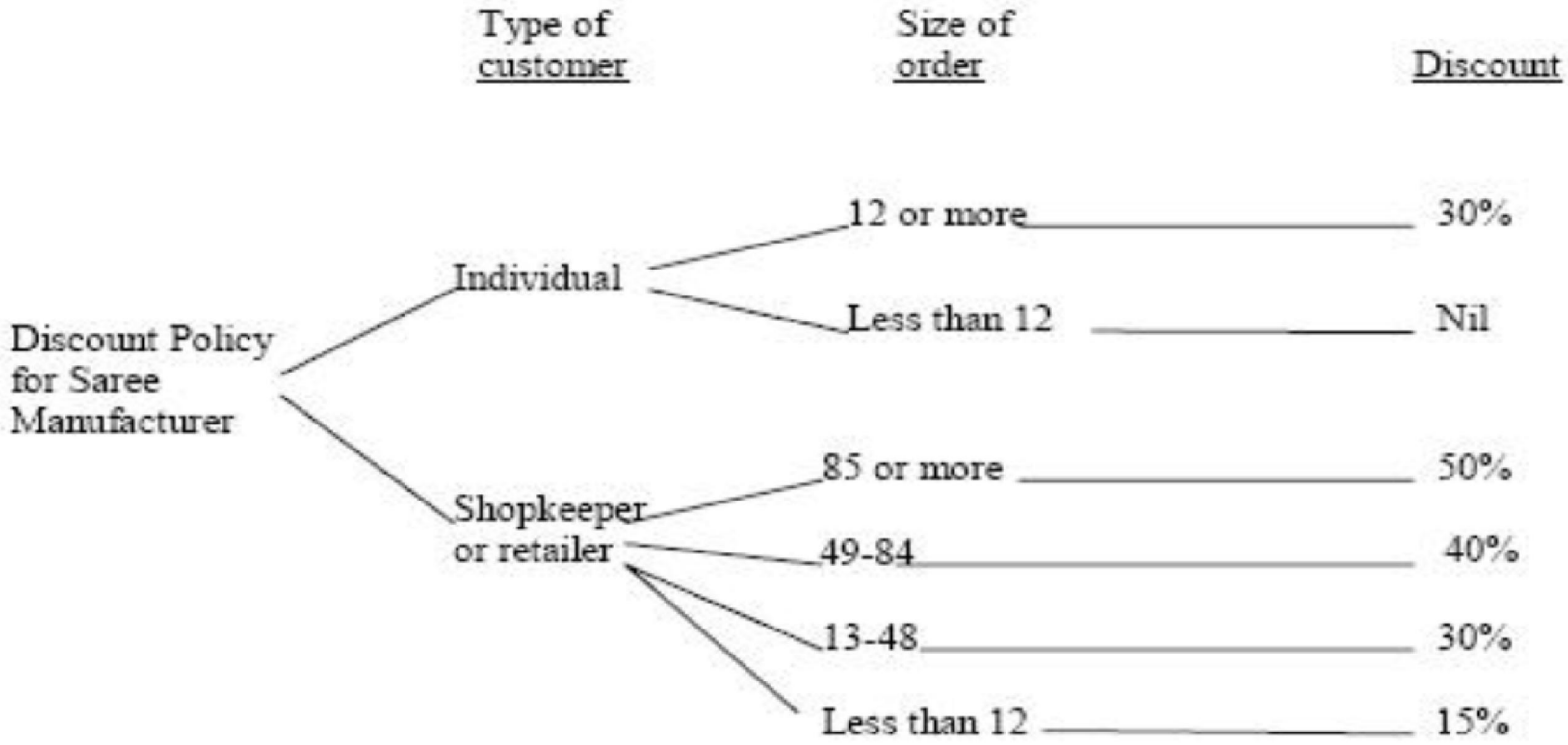


- In our day-to-day life, many a times we come across complex cases where the most appropriate action under several conditions is not apparent easily and for such a case a decision tree is a great aid. Hence this representation is very effective in describing the business problems involving more then one dimension and parameters.

Decision tree CONTD

- They also point out the required data, which surrounds the decision process. All the data used in the decision making should be first described and defined by the analyst so that the system can be designed to produce correct output data.
- **Example 1:** Consider the discount policy of a saree manufacturer for his customers.
- *"According to the policy the saree manufacturer give discount to his customers based on the type of customer and size of their order. For the individual, only if the order size is 12 or more, the manufacturer gives a discount of 30% and for less than 12 sarees the discount is not offered. Whereas in case of shopkeeper or retailers, the discount policy is different. If the order is less than 12 then there is 15% discount. For 13 to 48 sarees order, the discount is 30%, for 49 to 84 sarees the discount is 40% and for 85 sarees and more the discount is 50%".*

Decision tree CONTD



- The decision trees are not always the most appropriate and the best tool for the decision making process. Representing a very complex system with this tool may lead to a huge number of branches with a similar number of possible paths and options.
- For a complex problem, analyzing various situations is very difficult and can confuse the analyst

Decision tree CONTD

Example 2: Consider the recruitment policy of ABC Software Ltd.

- *"If the applicant is a BE then recruit otherwise not. If the person is from Computer Science, put him/her in the software development department and if the person is from non-computer science background put him/her in HR department. If the Person is from Computer Science and having experience equal to or greater than three years, take him/her as Team leader and if the experience is less than that then take the person as Team member. If the person recruited is from non Computer Science background, having experience less than three years, make him/her Management Trainee otherwise Manager"*

2. Decision tables

- A decision table is a table with various conditions and their corresponding actions. Decision table is a two dimensional matrix. It is divided into four parts, *condition stub, action stub, condition entry, and action entry* as shown in the figure below:
- **Condition stub** shows the various possible conditions.
- **Condition entry** is used for specifying which condition is being analyzed.
- **Action stub** shows the various actions taken against different conditions.
- And **action entry** is used to find out which action is taken corresponding to a particular set of conditions. The steps to be taken for a certain possible condition are listed by action statements. Action entries display what specific actions to be undertaken when selected conditions or combinations of conditions are true. At times notes are added below the table to indicate when to use the table or to distinguish it from other decisions tables.
- The right side columns of the table link conditions and actions and form the decision rules hence they state the conditions that must be fulfilled for a particular set of actions to be taken.
- In the decision trees, a fixed ordered sequence is followed in which conditions are examined. But this is not the case here as the decision rule incorporates all the required conditions, which must be true.

Decision tables CONTD

Condition stub	Condition entry
Action stub	Action entry

Developing Decision Tables

- Before describing the steps involved in building the decision table it is important to take a note of few important points.
- Every decision should be given a name and the logic of the decision table is independent of the sequence in which condition rules are written but the action takes place in the order in which events occur. Wherever possible, duplication of terms and meaning should be avoided and only the standardized language must be used. The steps of building the concerned tables are given below.

Decision tables CONTD

1. Firstly figure out the most essential factors to be considered in making a decision.

This will identify the conditions involved in the decision. Only those conditions should be selected which have the potential to either occur or not but partial occurrences are not permissible.

2. Determine the most possible steps that can take place under varying conditions and not just under current condition. This step will identify the actions.

3. Calculate all the possible combinations of conditions.

For every N number of conditions there are 2*2*2…. (N times) combinations to be considered.

4. Fill the decision rules in the table.

condition entries in a decision table are filled as Y/N and action entries are generally marked as "X". For the conditions that are immaterial a hyphen "-" is generally put. Decision table is further simplified by eliminating and consolidating certain rules. Impossible rules are eliminated. There are certain conditions whose values do not affect the decision and always result in the same action. These rules can be consolidated into a single rule.

Decision tables CONTD

Example 3: Consider the procedure for becoming a member of library in Noida library.

- *"Anyone whose age is 18yrs or more than that can become a member of library. There are two types of memberships depending upon the duration of membership. First is for 6 months and other is for 1 year. 6 months membership fee is Rs 500 and 1 year membership fee is Rs 1000".*
- **Example 2:** Consider the recruitment policy of ABC Software Ltd.
- **Example 1:** Consider the discount policy of a saree manufacturer for his customers.

Decision tables CONTD

Condition Stub	Condition Entry							
	1	2	3	4	5	6	7	8
Is person BE ?	Y	N	Y	N	Y	N	Y	N
Is person Comp Sci. Grad CS ?	Y	Y	N	N	Y	Y	N	N
Work Experience >= 3	Y	Y	Y	Y	N	N	N	N
Recruit	x		x		x		x	
Don't Recruit		x		x		x		x
S/W Deptt	x				x			
HR Deptt			x				x	
TeamLeader	x							
TeamMem					x			
MgtTrainee							x	
Manager			x					

Action Stub Action Entry

Decision tables CONTD

- The simplified table of the above figure is displayed in figure below

BE ?	Y	Y	Y	Y	N
CS?	Y	N	Y	N	-
WEX >= 3	Y	Y	N	N	-
Recruit	Y	Y	Y	Y	
Don't recruit					X
S/W Deptt	Y		Y		
HR Deptt		Y		Y	
TeamLeader	Y				
TeamMem			Y		
MgtTrainee				Y	
Manager		Y			

Decision tables CONTD

Condition Stub	Condition Entry					
	1	2	3	4	5	6
Customer is individual ?	Y	Y				
Customer shopkeeper or retailer ?			Y	Y	Y	Y
Order-size 85 copies or more ?			Y			
Order-size 49-84 sarees ?				Y		
Order-size 13-48 copies ?					Y	
Order-size 12 or more ?		Y				
Order-size less than 12?	Y					Y
Allow 50% discount		X	X			
Allow 40% discount				X		
Allow 30% discount	X				X	
Allow 15% discount						X

Decision table-Discount Policy

3. Structured English

- Structured English is one more tool available to the analyst. It comes as an aid against the problems of ambiguous language in stating condition and actions in decisions and procedures. Here no trees or tables are employed, rather with ***narrative statements*** a procedure is described. Thus it does not show but states the decision rules.
- The analyst is first required to identify the conditions that occur in the process, subsequent decisions, which are to be made and the alternative actions to be taken.
- Here the steps are clearly listed in the order in which they should be taken. There are no special symbols or formats involved unlike in the case of decision trees and tables, also the entire procedure can be stated quickly as only English like statements are used.
- Structured English borrows heavily from structured programming as it uses logical construction and imperative statements designed to carry out instructions for actions. Using "IF", "THEN", "ELSE" and "So" statement decisions are made. In this structured description terms from the data dictionary are widely used which makes the description compact and straight.

3. Structured English

Developing Structured Statements

Three basic types of statements are employed to describe the process.

1. Sequence Structures - A sequence structure is a single step or action included in a process. It is independent of the existence of any condition and when encountered it is always taken. Usually numerous such instructions are used together to describe a process.

2. Decision Structures - Here action sequences described are often included within decision structures that identify conditions. Therefore these structures occur when two or more actions can be taken as per the value of a specific condition. Once the condition is determined the actions are unconditional.

3. Iteration Structures- these are those structures, which are repeated, in routing operations such as DO WHILE statements.

Structured English CONTD

Example: Consider the discount policy of a saree manufacturer for his customers.

```
COMPUTE_DISCOUNT
See Number of Sarees
IF order is from individual
        and-if order is for 12 or more sarees
                THEN : Discount is 50%
        ELSE order is for fewer than 12 sarees
                SO: Discount is 30%
ELSE order is from shopkeeper or retailer
        SO-IF order is for 85 sarees or more
                Discount is 50%
            ELSE IF order is for 49 to 84 sarees
                        Discount is 40%
            ELSE IF order is for 48 to 13 sarees
                        Discount is 30%
            ELSE  order is for less than 12 sarees
             SO:    Discount is 15%
```

4. Data Dictionary

- As the name suggests the data dictionary is a **catalog or repository of data terms such as data elements, data structures** etc. *Data dictionary is* **a collection of data to be captured and stored in the system, inputs to the systems and outputs generated by the systems.** So let's first know more about what are these data elements and structures.

Data element: The smallest unit of data, which can not be further decomposed, is known as a data element. For example any number digit or an alphabet will qualify to be data elements. Data element is the data at the most fundamental level. These elements are used to as building blocks for all other data in the system. At times data elements are also referred as data item, elementary item or just as field. There is a very little chance that only by them data element can convey some meaning.

Data structure: Data elements when clubbed together as a group make up a data structure. These data elements are related to one another and together they stand for some meaning. Data structures are used to define or describe the system's components.

- Data dictionary entries consist of a set of details about the data used or produced in the system such as data flows, data stores and processes. For each item the dictionary records its name, description, alias and its length. The data dictionary takes its shape during the data flow analysis and its contents are used even till the system design. It very reasonable to know why the data dictionary is so essential. There are numerous important reasons.

Data Dictionary CONTD

- In a system there is data volume flow in the form of reports, documents etc. In these transactions either the existing data is used or new data items are created. This poses a potential problem for the analyst and thus developing and using a well-documented dictionary can be a great help.
- Now consider a case where everyone concerned with the system derives different meanings for the same data items. This problem can continue until the meaning of all data items and others are well documented so that everyone can refer to the same source and derive the same common meaning.
- Documentation in data dictionary is further carried on to record about the circumstances of the various process of the system. A data dictionary is always an added advantage for the system analysis. From the data dictionary one can determine about the need of the new features or about the changes required. Thus they help in evaluating the system and in locating the errors about the system description, which takes place when the contents of the dictionary are themselves not up to the mark.

CASE Tools

- ***CASE (Computer Aided Software Engineering)***
- ***A tool*** is any device, object or a kind of an operation used to achieve a specific task. The complete and correct description of the system is as important as the system itself. The analyst uses *CASE tool* to represent and assemble all the information and the data gathered about the system.
- Most of the organizations have to follow some kind of procedures and they are required to make all sorts of decisions from time to time. For the job of the analyst both these procedures and decision making processes of the business system under investigation are equally important.
- *Expressing business processes and rules in plain text is very cumbersome and difficult process*. It requires a lot of effort. Moreover, it does not guarantee if the reader will understand it well.
- So representing these things graphically is a good choice. So CASE Tools are useful in representing business rules and procedures of organization in graphical way. It requires less effort and it is easy to understand.

CASE Tools CONTD

- Some CASE tools are designed for creating new applications and not for maintaining or enhancing existing ones hence if an organization is in a maintenance mode, it must have a CASE tool that supports the maintenance aspect of the software developed. Many a times the large projects are too big to be handled by a single analyst thus here the CASE tool must be compatible enough to allow for partitioning of the project.
- Efficient and better CASE tools collect a wide range of facts, diagrams, and rules, report layouts and screen designs. The CASE tool must format the collected data into a meaningful document ready to use.
- Tools for DFD or a data flow diagram is a perfect example of a good CASE tool